



REACT 2026 DATATHON · TABULAR FRAUD DETECTION

Overfit & Overcaffeinated

Catching fraud that moves — without ever seeing the future.

Causal feature engineering → Dual-horizon LightGBM + Tabular ResNet → Temporal propagation

3.4×

Baseline PR-AUC

0

External data or pretrained models

Fraud is rare. And it keeps moving.

That combination — extreme imbalance plus a moving target — is what makes this a hard problem, not just a big one.

1.76%

of transactions are fraud — a needle-in-a-haystack label rate.

731,942

labeled training transactions, ordered strictly by time.

262,648

held-out test transactions, scored completely blind.

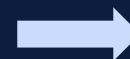


A regime shift, mid-competition

Our own audits showed summer attacks abandoning old patterns for new ones:

Winter pattern

Large, suspicious one-off amounts



Summer pattern

Fast device-velocity bursts

A model tuned only on early data would miss exactly the attacks that matter most by the time it ships — which is exactly why a 90-day specialist horizon exists in our approach.

Three stages. Zero shortcuts.

Every stage is built from scratch on the competition data — no external sources, no pretrained weights of any kind.



Blend: $0.48 \times \text{Full-history} + 0.52 \times 90\text{-day} \rightarrow 0.88 \times \text{Trees} + 0.12 \times \text{ResNet} \rightarrow \text{propagate over a } \pm 30\text{-min sliding window}$

Pretrained models used: **none.** 158 numerical + 2 categorical features · CPU-only training · ~15.5 min end-to-end (933s)

We trusted only what beat the noise.

Rigor cuts both ways: it's how we found real gains, and how we caught ourselves fooling ourselves.



Validation discipline

- **Chronological holdout**

Trained through June 30, tested on July 1-15 (59,465 txns) — never shuffled, mirroring real deployment.

- **Measured our own noise floor**

Seed variance ≈ 0.0020 PR-AUC. Only kept changes that beat it — not every wiggle is a win.

- **Zero-leakage guarantee**

Every feature checked by an automated assert against using future information.



What we deliberately rejected

- **Forward-looking features**

+0.024 local lift — but broke strict temporal-causality rules, so we cut them entirely.

- **Naive class-reweighting**

scale_pos_weight distorted ranking under PR-AUC. Removing it was worth +0.026 on the leaderboard.

Being wrong quickly, on purpose, is how the real gains stayed real.



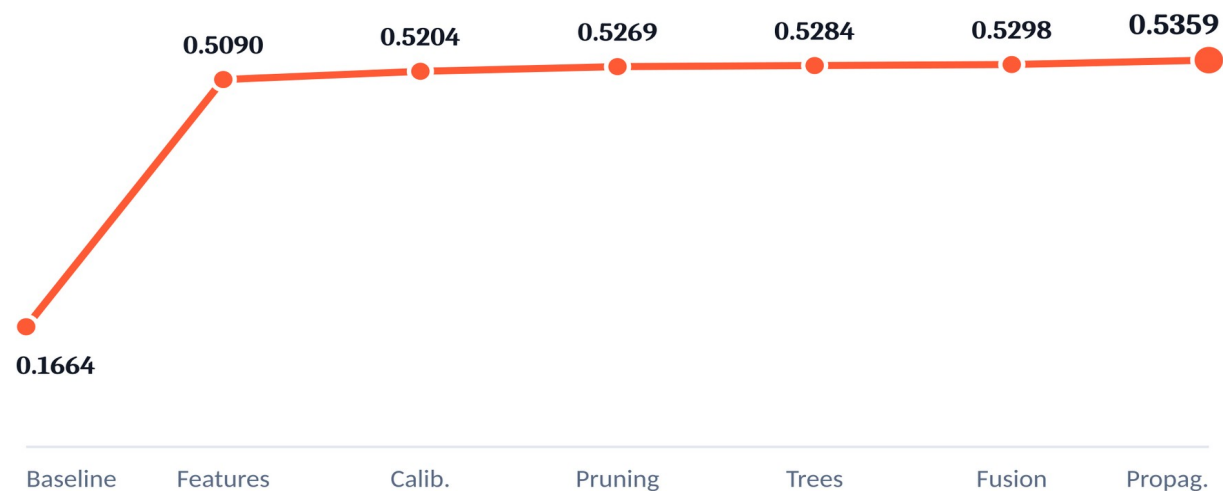
Preprocessing & feature engineering: 220+ engineered signals — expanding stats, burst ratios, novelty flags, graph components — pruned to 160 by information density. Full breakdown in the appendix.

RESULTS

0.56548 Public LB — 3.4× the baseline.

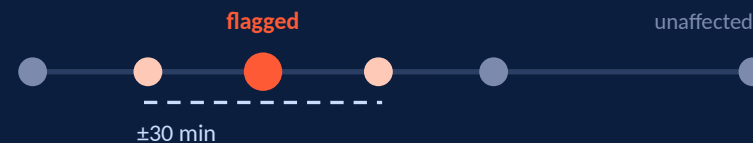
Every stage of the pipeline earned its place on the scoreboard.

Local PR-AUC by milestone



This notebook's own last run reproduced 0.5341 local PR-AUC (vs. 0.5359 reported) — normal LightGBM thread-level variance, not a different pipeline.

How propagation behaves



A confirmed signal lifts only its ±30-min neighbors — 4.7% of rows. The other 95.3% pass through untouched.

0.56548

Public Leaderboard · rank stated inconsistently across our own materials — confirm before presenting

Sibling fraud is 3.9× more likely on the same device, 2.6× on the same customer — the pattern propagation exploits.



IMPACT & CLOSE

Fraud doesn't hold still. Neither should the model that catches it.

Causally-safe features and burst-aware propagation aren't just leaderboard tricks — the same design could run in a live payment pipeline, adapting to new attack patterns without waiting for a retrain.

Thank you — happy to take questions.

0.56548

Public Leaderboard

3.4×

the baseline PR-AUC

~15.5 min

CPU-only, end-to-end



APPENDIX

Technical deep-dive — backup detail for Q&A

Not part of the timed 3-minute talk — reference only if judges want more.

160 features. All strictly past-only.

Pruned from 220+ candidates by information density. Every feature passes an automated point-in-time leakage assert.

220+ candidate features → pruned by information density → 160 kept in the final model



Scale-free deviations

```
cust_amt_ratio  
cust_amt_robust_z
```

Amount normalized against each customer's own expanding history, not raw dollars. `cust_amt_ratio` alone is the single highest-gain feature.



Trailing windows

```
5m · 15m · 30m · 1h · 3h ·  
6h · 12h · 24h · 72h · 168h
```

Sub-hour, traffic-matched windows catch automated credential-stuffing bursts on high-throughput devices and merchants.



Burst acceleration

```
cust_burst_accel =  
(cnt1h×24)/(cnt24h+1)
```

Normalized velocity ratios detect abrupt spikes relative to a customer's own recent baseline.



Counterparty novelty

```
is_new_device_for_customer  
is_new_location_  
for_customer
```

Real-time account-takeover signals from device and location novelty, plus shared-device concentration.



Graph connectivity

```
cd_device_id_component_size  
cm_customer_id_  
component_size
```

Union-Find over customer ↔ device and customer ↔ merchant graphs — a growing shared component flags fraud rings before they strike again.

Two very different models, one blend.

Trees capture sharp categorical rules; the neural net smooths the boundaries between them.



Dual-Horizon LightGBM

Full-Train Horizon *Jan 1 – Jul 15*

882 rounds · 127 leaves · lr 0.02 · feat.frac 0.85 · min_leaf 50

90-Day Specialist *Apr 16 – Jul 15*

433 rounds · 63 leaves · lr 0.02 · feat.frac 0.75 · min_leaf 100

Blend: $0.48 \times \text{Full-Train} + 0.52 \times \text{90-Day} \rightarrow 0.5284 \text{ local PR-AUC}$



Tabular ResNet

Numeric + categorical input



StandardScaler (clipped) + entity embeddings



3 × Residual Block — LayerNorm → Linear(256,512) → GELU → Dropout(0.15) → Linear(512,256) → Dropout(0.15) → + Skip



Output

Training: AdamW (lr 1e-3, wd 1e-4) · CosineAnnealingLR · 8 epochs · CPU-only

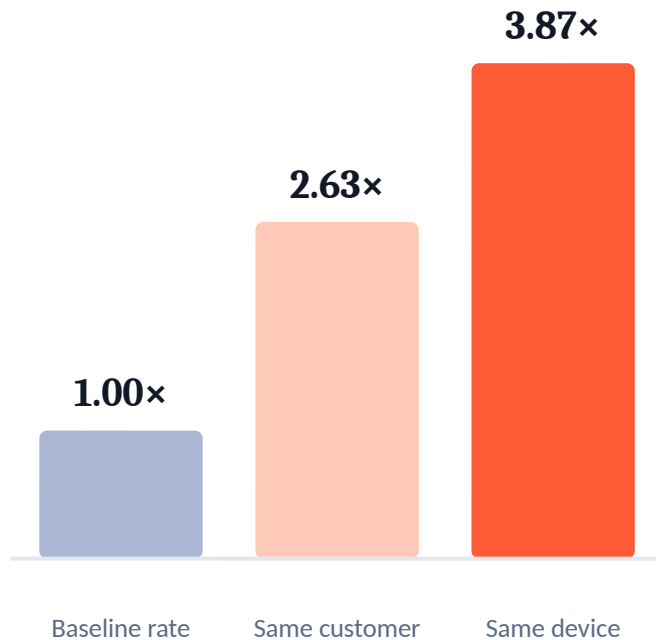
Blend: $0.88 \times \text{Trees} + 0.12 \times \text{ResNet} \rightarrow 0.5298 \text{ local PR-AUC}$

Why propagation works: fraud has siblings.

A confirmed fraud sharply raises the odds its near-in-time neighbors are fraud too.



P(sibling is fraud | current is fraud)



The propagation rule

$$P_{\text{final}}(i) = (1 - w) \cdot P_{\text{raw}}(i) + w \cdot \text{avg}(PL00, \text{device}(i), PL00, \text{customer}(i))$$

$$w = 0.50 \cdot \text{sliding } \pm 30\text{-minute window (60-min span)}$$

- Singleton transactions with no in-window neighbor (95.3% of test rows) pass through completely unchanged.
- Only the 4.7% of transactions sitting inside a real time-cluster get their score adjusted.
- The operation touches zero labels and fits zero test-time parameters — purely structural, not learned on the test set.

The full milestone ledger.

Every technique that shipped, and exactly what it was worth — locally and on the public leaderboard.

#	Milestone	What changed	Local PR-AUC	Public LB
0	Baseline	Raw tabular features, default LightGBM	0.1664	0.1664
1	Feature Pipeline	Expanding baselines, robust MAD z-scores, trailing windows	0.5090	0.5105
2	Objective Calibration	Removed naive scale_pos_weight class reweighting	0.5204	0.5395
3	Feature Pruning	Information-density pruning to the top 160 core features	0.5269	—
4	Dual-Horizon Trees	48% full-train + 52% 90-day specialist LightGBM	0.5284	—
5	Tree-Neural Fusion	88% dual-horizon LightGBM + 12% Tabular ResNet	0.5298	—
6	Temporal Propagation	Sliding LOO propagation over a ± 30 -min window	0.5359	0.56548

Dashes mark milestones evaluated only on the local chronological holdout, not resubmitted individually to the leaderboard.